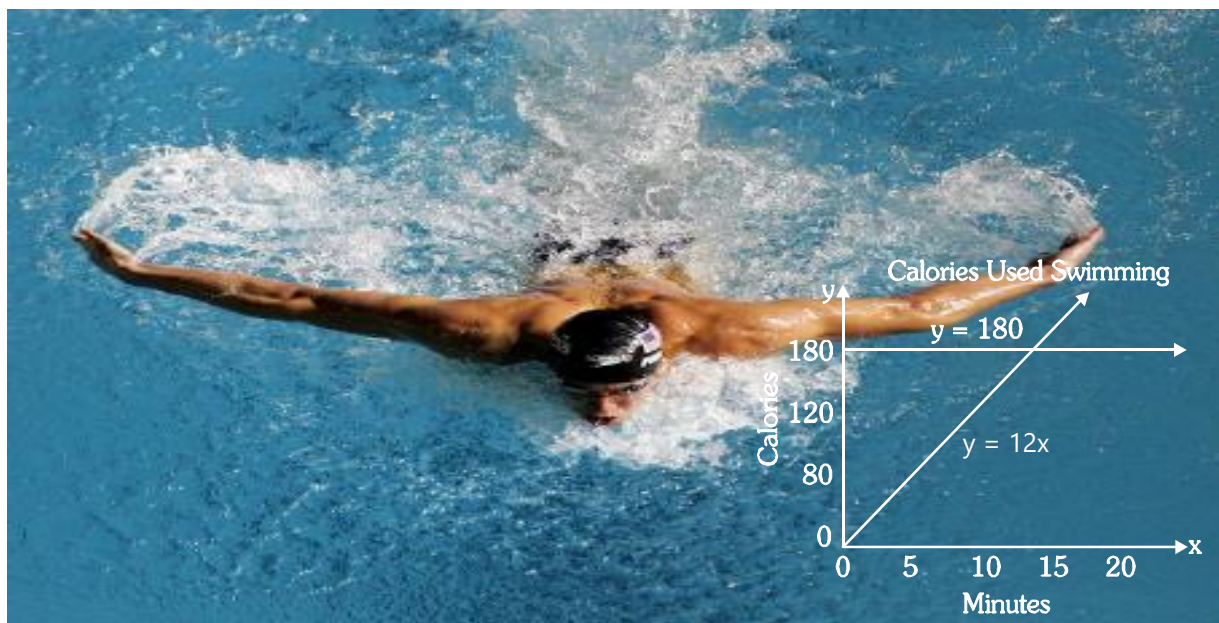


7

Linear Equations in two Variables



Linear equation

An equation in which the maximum power on variable is one, is called a linear equation.

For e.g. $4x + 5 = 3x + 1$, $2x + 3y = 45$

Solution of a linear equation in one variable

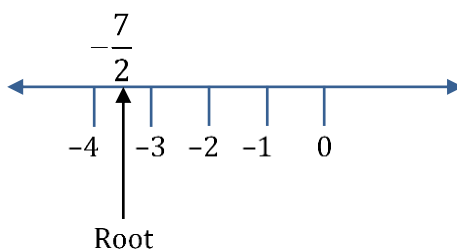
Value of the variable which when substituted in equation, makes two sides of the equation equal, is called the solution of the equation.

For e.g. for linear equation $ax + b = 0$, the solution is $-\frac{b}{a}$ because if we substitute $-\frac{b}{a}$, for x

in $ax + b = 0$, the two sides of the equation become equal.

For e.g. Consider the equation $2x + 7 = 0$.

Its solution i.e. the root of the equation is $-\frac{7}{2}$. This can be represented on the number line as shown in the figure.



**Quick
Tips**

The solution of a linear equation is not affected when :

- ★ The same number is added or subtracted from both the sides of the equation.
- ★ We multiply or divide both sides of equation by the same non-zero number.

**Do You
Remember ?**

- ★ In our earlier classes we have studied about linear equations in one variable, their formation, solving methods. In this chapter we will extend our knowledge about equation by introducing 2 variables in the linear equation.

Linear equation in two variables

An equation which can be expressed in the form $ax + by + c = 0$, where a , b and c are real numbers such that a and b are not zero, is called a linear equation in two variables x and y . In the linear equation $ax + by + c = 0$, a is coefficient of x , b is coefficient of y and c is constant. Here a , b , c are called the coefficients of the linear equation.

**Building
Concepts****1**

Express the equation $7x = 3y$ in the form of $ax + by + c = 0$ and find the coefficients of linear equation.

Explanation

$7x = 3y$ can be written as $7x - 3y = 0$

By comparing this equation with $ax + by + c = 0$

coefficient of $x = a = 7$

coefficient of $y = b = -3$

constant term = $c = 0$

$\therefore a = 7, b = -3, c = 0.$



A linear equation in two variables has infinitely many solutions.

SPOT LIGHT**Check your
Concepts****1**

Write each of the following equations as equations in two variables ($ax + by + c = 0$) and find the value of a , b and c .

(i) $x = 8$

(ii) $y = -3$

(iii) $x - 3 = \sqrt{5}y$

(iv) $2 = 5x - 3y$



**Building
Concepts**

2

In a one-day international cricket match between India and Australia played in Kolkata, Dhoni and Yuvraj together scored 198 runs. Express this information in the form of an equation.

Explanation

Let the runs scored by Dhoni be x and the runs scored by Yuvraj be y .

Dhoni and Yuvraj together scored 198 runs.

So, $x + y = 198$

$\Rightarrow x + y - 198 = 0$ is the required equation.

Solution of linear equation in two variables

Solution of linear equation in two variables i.e. $ax + by + c = 0$ has a pair of values, one for the unknown variable x and one for the second unknown variable y which satisfy the given equation.

Let us consider the equation $2x + 3y = 6$. When we substitute $x = 1$, $y = \frac{4}{3}$ in the given

equation, we find that $2 \times 1 + \frac{4}{3} \times 3 = 2 + 4 = 6$.

i.e. the equation is satisfied for the pair of values $x = 1$ and $y = \frac{4}{3}$.

Therefore $(1, \frac{4}{3})$ is a solution of the linear equation.

We can find some more solutions of this equation like $(2, \frac{2}{3})$, $(0, 2)$, $(3, 0)$, $(4, \frac{-2}{3})$ and so

on i.e. infinitely many solutions. So a linear equation in two variables has infinitely many solutions.



**Building
Concepts**

3

Which of the following equations have a unique solution or infinitely many solutions.

(i) $x + 1 = 6$

(ii) $y = 3x + 2$

Explanation

(i) The given equation is $x + 1 = 6$

$\Rightarrow x = 6 - 1$

$\Rightarrow x = 5$

$\Rightarrow x + 0.y - 5 = 0, \quad \dots (i)$

Thus, $x = 5$ and any value of y is a solution.

Hence, equation (i) has infinitely many solution

(ii) The given equation is $y = 3x + 2$

$$\therefore 3x + (-1)y + 2 = 0 \quad \dots (ii)$$

which is a linear equation in two variable.

Hence, equation (ii) has infinitely many solutions



If (1, 3) is a solution of the equation $3x + 5y = b$, then find the value of b.

Explanation

As (1, 3) is a solution of the given equation.

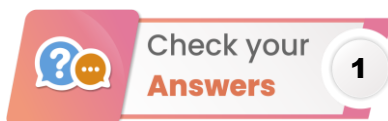
$\therefore x = 1, y = 3$ satisfies the given equation.

Then, the equation $3x + 5y = b$ is reduced to

$$3 \times 1 + 5 \times 3 = b$$

$$\Rightarrow 3 + 15 = b$$

$$\Rightarrow b = 18$$



(i) $a = 1, b = 0, c = -8$

(ii) $a = 0, b = 1, c = 3$

(iii) $a = 1, b = -\sqrt{5}$ and $c = -3$

(iv) $a = 5, b = -3$ and $c = -2$



Show that $(x = 1, y = 1)$ as well as $(x = 2, y = 5)$ is a solution of $4x - y - 3 = 0$.

Solution

If we put $x = 1$ and $y = 1$ in the given equation,

we have L.H.S. $= 4 \times 1 - 1 - 3 = 0 = \text{R.H.S.}$

so, $x = 1, y = 1$ is a solution of $4x - y - 3 = 0$

If we put $x = 2, y = 5$ in the equation $4x - y - 3 = 0$, we have

L.H.S. $= 4 \times 2 - 5 - 3 = 0 = \text{R.H.S.}$

So, $x = 2, y = 5$ is a solution of the equation $4x - y - 3 = 0$



Check which of the following are solutions of the equation $x - 2y = 4$ and which are not :

- (i) (0, 2) (ii) (4, 0) (iii) $(\sqrt{2}, 4\sqrt{2})$



Find two solutions of the equation : $4x + 3y = 12$

Solution

The given equation is $4x + 3y = 12$... (1)

Putting $x = 0$ in (1), we get

$$4 \cdot 0 + 3y = 12$$

$$\Rightarrow 3y = 12$$

$$\Rightarrow y = 4$$

$$\therefore x = 0 \text{ and } y = 4$$

So, (0, 4) is a solution of the given equation (1).

Further, putting $y = 0$ in (1), we get

$$4x + 3 \cdot 0 = 12$$

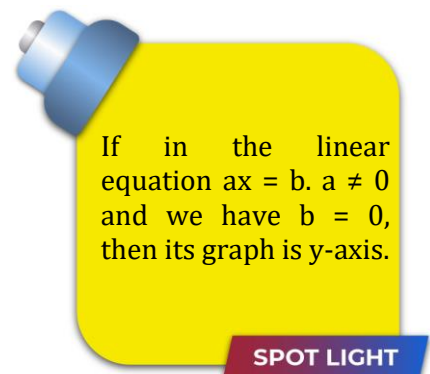
$$\Rightarrow 4x = 12$$

$$\Rightarrow x = 3$$

$$\therefore x = 3 \text{ and } y = 0$$

So, (3, 0) is a solution of the given equation (1).

Hence, (0, 4) and (3, 0) are two solutions of the given equation.



- ★ While converting a given equation to the standard form $ax + by + c = 0$ and finding values of a , b & c , students can find different answers depending on the transposition they are doing.

Eg. $-2x + 3y = 6$

$$\Rightarrow -2x + 3y - 6 = 0$$

$$a = -2, b = 3, c = -6 \quad (\text{correct})$$

$$\Rightarrow 2x - 3y + 6 = 0$$

$$a = 2, b = -3, c = 6 \quad (\text{correct})$$

★ Transposition rules should be very clear to us. Following are the common mistakes.

For e.g.

(i) $2x = 0$

$\Rightarrow x = -2$ (wrong)

$\Rightarrow x = \frac{0}{2} = 0$ (correct)

(ii) $-5x = 10$

$\Rightarrow x = \frac{10}{5} = 2$ (wrong)

$\Rightarrow x = \frac{10}{-5} = -2$ (correct)

(iii) $2x + 3 = 4$

$\Rightarrow x + 3 = \frac{4}{2}$ (wrong)

$\Rightarrow 2x = 4 - 3$

$\Rightarrow 2x = 1$

$\Rightarrow x = \frac{1}{2}$ (correct)

Graph of a linear equation in two variables

In order to draw the graph of a linear equation $ax + by + c = 0$, $a \neq 0$, $b \neq 0$, we may follow the following algorithm.

Step-I : Obtain the linear equation and let the equation be $ax + by + c = 0$.

Step-II : Express y in terms of x to obtain $y = -\left(\frac{ax + c}{b}\right)$

or x in terms of y to obtain $x = -\left(\frac{by + c}{a}\right)$

Step-III : Put any two or three values of x and calculate the corresponding values of y from the expression in step-II to obtain two solution say (a_1, b_1) and (a_2, b_2) , if possible take values of x as integers in such a manner that the corresponding values of y are also integers.

Step-IV : Plot points (a_1, b_1) and (a_2, b_2) .

Step-V : Join the points marked in step IV to obtain a line.

The line obtained is the graph of the equation $ax + by + c = 0$.



Draw the graph of the equation $3x - 2y = 7$.

Explanation

We have, $3x - 2y = 7 \Rightarrow x = \frac{7+2y}{3}$

When $y = 1$, $x = \frac{7+2}{3} = 3$

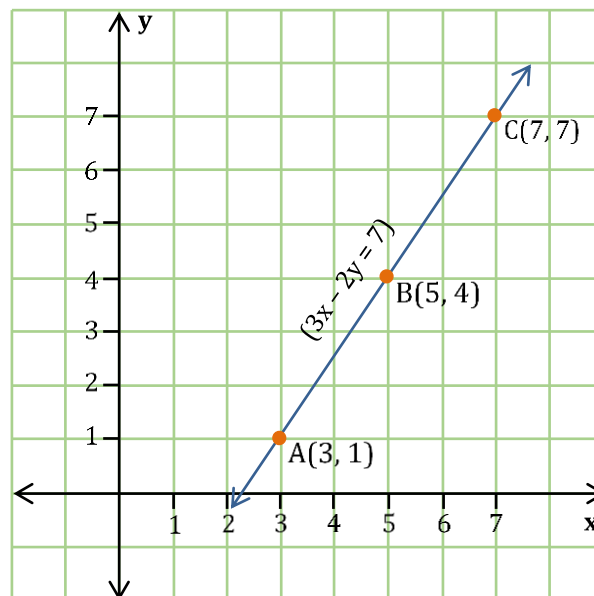
When $y = 4$, $x = \frac{7+8}{3} = 5$

When $y = 7$, $x = \frac{7+14}{3} = 7$

Thus, we have the following table exhibiting the abscissa and ordinates of the points on the line represented by the given equation.

x	3	5	7
y	1	4	7

Plotting the points A(3, 1), B(5, 4) and C(7, 7) on the graph paper and joining the points A, B and C, we get a straight line.



If in the linear equation $ay = b$, $a \neq 0$ and we have $b = 0$, then its graph is x-axis

SPOT LIGHT



Be Alert !

- ★ Although only 2 points (solutions) are required to draw graph of a linear equation in 2 variables but it is advisable to find 3 solutions or points so that if there is any mistake in finding the points or plotting then we will come to know about this while plotting the points.



- (i) $(0, 2)$ is not a solution of the given equation.
 (ii) $(4, 0)$ is a solution of the given equation.
 (iii) $(\sqrt{2}, 4\sqrt{2})$ is not a solution of the given equation.

Equations of Lines Parallel to the x-axis and y-axis

$x = a$ is an equation of line parallel to y-axis and $y = b$ is an equation of line parallel to x – axis.



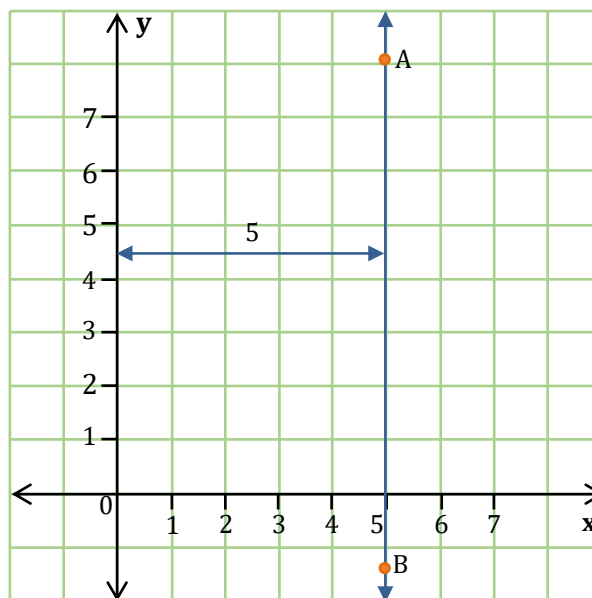
Represent $x - 5 = 0$ on the Cartesian plane.

Solution

We solve $x - 5 = 0$

$$x = 5$$

Here, the graph AB is line parallel to the y-axis and at a distance of 5 units to the right of it.



Equation of line parallel to x-axis and y-axis:

- ★ Equations which doesn't have any term of x will represent a straight line parallel to x-axis.
For e.g. $y = 3, y = -5$ etc.
- ★ Equation which doesn't have any term of y will represent a straight line parallel to y-axis.
For e.g. $x = 1, x = -3$ etc.

Memory map

Linear Equations

An equation in which the power of variable is one, is called a linear equation.

eg. $4x + 5 = 3x + 1$, $2x + 3y = 45$

Solution of a Linear Equation in one Variable

Value of the variable which when substituted in equation, makes two sides of the equation equal, is called the solution of the equation.

For e.g. Consider the equation $2x + 7 = 0$ its solution i.e. the root of the equation is $-\frac{7}{2}$.

The solution of a linear equation is not affected when :

- (i) The same number is added or subtracted from both the sides of the equation.
- (ii) We multiply or divide both sides of equation by the same non-zero number.

Linear Equation in Two Variables

An equation which can be expressed in the form $ax + by + c = 0$, where a , b and c are real numbers such that a and b are not zero, is called a linear equation in two variables x and y .

In the linear equation $ax + by + c = 0$, a is coefficient of x , b is coefficient of y and c is constant. Here a , b , c are called the coefficients of the linear equation.

Solution of Linear Equation in Two Variables

Solution of linear equation in two variables i.e. $ax + by + c = 0$ has a pair of values, one for the unknown variable x and one for the second unknown variable y which satisfy the given equation.

For e.g. consider the equation $2x + 3y = 6$. When we substitute $x = 1$, $y = \frac{4}{3}$ in the given

equation, we find that $2 \times 1 + \frac{4}{3} \times 3 = 2 + 4 = 6$ i.e. the equation is satisfied for the pair of

values $x = 1$ and $y = \frac{4}{3}$. Therefore $(1, \frac{4}{3})$ is a solution of the linear equation.

A linear equation in two variables has infinitely many solutions.

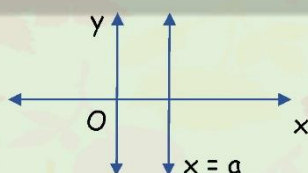
Graph of a Linear Equation in Two Variables

In order to draw the graph of a linear equation $ax + by + c = 0$, $a \neq 0$, $b \neq 0$, we may follow the following algorithm.

-  **Step-I :** Obtain the linear equation and let the equation be $ax + by + c = 0$.
-  **Step-II :** Express y in terms of x to obtain $y = -\left(\frac{ax + c}{b}\right)$
-  **Step-III :** Put any two or three values of x and calculate the corresponding values of y from the expression in step-II to obtain two solution say (a_1, b_1) and (a_2, b_2) , if possible take values of x as integers in such a manner that the corresponding values of y are also integers.
-  **Step-IV :** Plot points (a_1, b_1) and (a_2, b_2) .
-  **Step-V :** Join the points marked in step IV to obtain a line.

The line obtained is the graph of the equation $ax + by + c = 0$.

Equations of Lines Parallel to the x -axis and y -axis



-  $x = a$ is an equation of line parallel to y -axis

-  $y = b$ is an equation of line parallel to x -axis.

